ACADEMIC CENTRAL

PRACTICE EXAMINATION

NC(V) LEVEL 4

ELECTRONICS AND DIGITAL CONCEPTS FOR ROBOTICS

PAPER 1 · PRACTICE D · MARKING GUIDELINE

MARKS: 100

Marking guideline for the Academic Central NC(V) Level 4 Electronics and Digital Concepts for Robotics paper 1 · practice d.

Written by Academic Central for practice, following the NC(V) subject guidelines for NC(V) Level 4. It is not a Department of Higher Education and Training marking guideline.

How to mark: each ✓ is one mark. Give marks for the formula, the substitution and the answer with the correct unit. Use consistent accuracy. Accept other correct answers and neat, recognisable diagrams.

SECTION A

QUESTION 1

1 1.1 B ✓ 1.2 D ✓ 1.3 A ✓ 1.4 C ✓ 1.5 C ✓ 1.6 A ✓ 1.7 D ✓ 1.8 B ✓ 1.9 A ✓ 1.10 C ✓ (10)

[10]

QUESTION 2

2 2.1 True ✓ 2.2 False ✓ 2.3 True ✓ 2.4 False ✓ 2.5 False ✓ 2.6 True ✓ 2.7 True ✓ 2.8 True ✓ 2.9 False ✓ 2.10 True ✓ (10)

[10]

QUESTION 3

3 3.1 F ✓ 3.2 B ✓ 3.3 A ✓ 3.4 E ✓ 3.5 C ✓ 3.6 J ✓ 3.7 H ✓ 3.8 I ✓ 3.9 D ✓ 3.10 G ✓ (10)

[10]

TOTAL SECTION A: [30]

SECTION B

QUESTION 4

4.1.1 $X_C = 1 \div (2\pi fC) = 1 \div (2\pi \times 50 \times 10 \times 10^{-6})$ ✓ $= 318{,}31\ \Omega$ ✓ (2)

4.1.2 $Z = \sqrt{(R^2 + X_C{}^2)}$ ✓ $= \sqrt{(220^2 + 318{,}31^2)}$ ✓ $= 386{,}94\ \Omega$ ✓ (3)

4.1.3 $I = V \div Z = 230 \div 386{,}94$ ✓ $= 0{,}59\ A$ ✓ (2)

4.2 An RC network (with a variable resistor) delays the voltage that reaches the DIAC in each half-cycle ✓; when the capacitor voltage reaches the DIAC's breakover voltage, the DIAC conducts and sends a pulse to the TRIAC gate ✓; the TRIAC then conducts for the rest of that half-cycle, so turning the variable resistor changes the firing point and therefore the brightness ✓. (3)

[10]

QUESTION 5

5.1 $V_2 = V_1 \times N_2 \div N_1 = 230 \times 115 \div 1\ 150$ ✓ $= 23\ V$ ✓ (2)

5.2 $P_{out} = 23 \times 2 = 46\ W$ ✓; $\eta = 46 \div 50 \times 100\%$ ✓ $= 92\%$ ✓ (3)

5.3 $E_T = 1{,}2\ V$ ✓; $r_T = 0{,}2 \div 4$ ✓ $= 0{,}05\ \Omega$ ✓; $I = 1{,}2 \div (1 + 0{,}05)$ ✓ $= 1{,}14\ A$ ✓ (5)

[10]

QUESTION 6

6.1 Monostable: one stable state; a trigger gives ONE output pulse of a set length, then it returns to the stable state ✓✓. Astable: no stable state; it switches continuously between high and low, producing a square wave without a trigger ✓✓. Bistable: two stable states; it stays in either state until a trigger changes it (like a flip-flop / memory) ✓✓. (6)

6.2 $t = 1{,}1 \times R \times C$ ✓ $= 1{,}1 \times 100\ 000 \times 22 \times 10^{-6}$ ✓ $= 2{,}42\ s$ ✓ (3)

6.3 Any ONE ✓: cleaning up a noisy sensor signal (line or light sensor) before a microcontroller input; switch debouncing; turning an analogue sensor voltage into a clean on/off signal without chatter. (1)

[10]

QUESTION 7

7.1 AC supply from a transformer ✓; four diodes in a correct bridge arrangement ✓; filter capacitor across the DC output with the correct polarity ✓; load resistor across the output, + and − labelled ✓ (4)

7.2 A bigger capacitor stores more charge ✓, so its voltage drops less between the peaks while it supplies the load, and the output stays closer to the peak value ✓. (2)

7.3 Any FOUR in a sensible order ✓✓✓✓: switch off and disconnect the supply (discharge the capacitors); identify the pins (input, ground, output) and note the orientation; desolder the faulty regulator with a solder sucker or wick without damaging the pads; fit the correct replacement (7805, same package) the right way round, with its heat sink and thermal paste if used; solder neatly and check for solder bridges; switch on and measure 5 V at the output before connecting the load. (4)

[10]

QUESTION 8

8.1 $V_p = 2 \times 5$ ✓ $= 10\ V$ ✓ (2)

8.2 $T = (1{,}5 + 2{,}5) \times 0{,}2$ ✓ $= 0{,}8\ ms$ ✓ (2)

8.3 $f = 1 \div T = 1 \div 0{,}0008$ ✓ $= 1\ 250\ Hz$ ✓ (2)

8.4 $\text{Duty cycle} = (1{,}5 \div 4) \times 100\%$ ✓ $= 37{,}5\%$ ✓ (2)

8.5 A ×10 probe divides the signal by 10 before it reaches the oscilloscope ✓ (so larger voltages can be measured with less loading of the circuit); if the oscilloscope is not set for ×10, the reading must be multiplied by 10 to get the real voltage ✓. (2)

[10]

QUESTION 9

9.1 All eight input combinations listed ✓; outputs ✓✓✓ (deduct 1 for each wrong output): (4)

A	B	C	F
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	1
1	0	0	0
1	0	1	1
1	1	0	1
1	1	1	1

9.2 Correct Karnaugh map with 1s in the four cells where C = 1 ✓; the four 1s grouped as one quad ✓; F = C ✓ (3)

9.3 NOT gate: one NAND gate with both inputs joined ✓. AND gate: a NAND gate ✓ followed by a second NAND gate with its inputs joined (used as a NOT gate) ✓. (3)

[10]

QUESTION 10

10.1 $n_s = (120 \times f) \div p = (120 \times 50) \div 4$ ✓ $= 1\,500$ r/min ✓; $s = ((1\,500 - 1\,440) \div 1\,500) \times 100\%$ ✓ $= 4\%$ ✓ (4)

10.2 The three-phase currents in the stator windings set up a magnetic field that rotates at synchronous speed ✓; the field cuts the rotor bars and induces currents in them ✓; these currents make their own magnetic field, which is pulled along by the stator field, so the rotor turns (slightly slower than the field) ✓. (3)

10.3 Any THREE ✓✓✓: no brushes or commutator, so less maintenance; strong and reliable (squirrel cage); cheaper; run directly from the three-phase supply; speed easily controlled with a VSD; nearly constant speed under load. (3)

[10]

TOTAL SECTION B: [70]

TOTAL: 100

END OF MARKING GUIDELINE